Uniform Laguerre signs and hypercomplex jets of the theta heat flow

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Abstract

For the theta heat family H, the two quadratic forms $H_x^2 - HH_{xx}$ and $H_{xx}^2 - H_xH_{xxx}$ control the evolution of its first jet at genuine stationary points. We prove that both forms are strictly positive, uniformly for $0 < t \le 1/2$ and sufficiently large $|x|$, whenever $t\log(|x|/(4\pi)) \ge 4 + \varepsilon$, for any fixed $\varepsilon > 0$. The proof derives a common nonvanishing complex amplitude for all four derivatives from Polymath's effective Riemann–Siegel approximation. Shrinking complex discs control derivative errors without differentiating a moving arithmetic cutoff. A quaternionic first jet and an explicit octonionic monogenic extension express the two signs as oriented associators. An exact common theta carrier and centered residual identities formulate the remaining inner-region estimate. Complete-source certificates establish both stationary signs and a physical first-jet gap through $|x| \le 104$ for $0 \le t \le 1/2$. The outer theorem has no evaluated height cutoff, and the inner small-time region remains unresolved.

Contents

1 The theta heat flow and the main result **2**
1.1 The fixed source and normalization . . . 2
1.2 Uniform positivity outside the inner region . . . 3

2 Proof of the uniform outer-region theorem **4**
2.1 A holomorphic comparison amplitude . . . 4
2.2 Uniform replacement of the weighted sums . . . 5
2.3 Complete value errors and derivative control . . . 6
2.4 The common phase and the two signs . . . 6

3 Why positive-time collisions are the obstruction **7**
3.1 A sufficient estimate only on two stationary sets . . . 8

4 The four jets as quaternionic and octonionic fields **9**
4.1 A quaternionic heat field with independent external time . . . 10
4.2 An eight-dimensional monogenic extension . . . 11
4.3 Oriented associators preserve the signed observations . . . 12
4.4 Why a source-specific estimate remains necessary . . . 13

5 Centered residuals and the common theta carrier **14**
5.1 A contour identity for the physical derivatives . . . 14
5.2 A residual adapted to genuine stationarity . . . 14
5.3 One carrier for all four derivatives . . . 15

6 The bounded region: a complete certificate through height 104 **16**
6.1 A central estimate with no theta truncation 17
6.2 What the interval certificates enclose 18
6.3 Coverage, the centered-residual pilot, and reproducibility 18

7 Obstructions and the remaining signed estimate **19**
7.1 Absolute first-channel dominance fails below four 19
7.2 The pointwise integrand need not have the desired sign 20
7.3 A small residual norm can impose an unnecessary condition 20

A Complete source and integration error bounds **21**

B Reproducibility and scope of the checks **22**

1 The theta heat flow and the main result

The four real derivatives

$$h_j(t,x) = \partial_x^j H(t,x), \qquad j = 0,1,2,3, \tag{1.1}$$

form the smallest consecutive jet containing the collision detector (H, H_x) and both quadratic expressions

$$\mathcal{L}_0 = h_1^2 - h_0 h_2, \qquad \mathcal{L}_1 = h_2^2 - h_1 h_3. \tag{1.2}$$

Here "four jets" means these four consecutive derivative components. Their common scalar origin is essential: they are not four independent fields. The first two give the quaternionic observation $F = h_0 + \mathbf{j}h_1$ and its energy $E = h_0^2 + h_1^2$. Since $H_t = -H_{xx}$,

$$\frac{1}{2}E_t = -h_0h_2 - h_1h_3 = \begin{cases} \mathcal{L}_0, & h_1 = 0, \\ \mathcal{L}_1, & h_2 = 0. \end{cases} \tag{1.3}$$

The stationary sets in this identity are those of the original scalar function and its first derivative. Their signs will remain the same objects throughout the analytic and hypercomplex constructions.

1.1 The fixed source and normalization

The family under consideration is

$$\Phi(u) = \sum_{n\geq 1} \phi_n(u), \qquad \phi_n(u) = (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{1.4}$$

$$H(t,z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \qquad H(0,z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right). \tag{1.5}$$

We use the convention $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$, agreeing with [3]. In particular the critical line corresponds to real z, and the heat equation has the backward sign

$$\partial_t H = -\partial_z^2 H. \tag{1.6}$$

For completeness, the normalization follows directly from the theta Mellin identity. Put $A(u) = e^u \sum_{n\geq 1} e^{-\pi n^2 e^{4u}}$. Poisson summation makes $P(u) = e^u + 2A(u)$ even, and differentiation

gives $\Phi = (P'' - P)/16 = (A'' - A)/8$. Splitting the theta Mellin integral at one and substituting $y = e^{4u}$ gives

$$\xi(1/2 + iz/2) = \frac{1}{2} - (1 + z^2) \int_0^\infty A(u) \cos(zu)\, du.$$

Evenness of P gives $A'(0) = -1/2$. Two integrations by parts now prove (1.5); all positive-infinity boundary terms vanish. This also proves that the complete source Φ is even. For $u \geq 0$ its terms are positive, and their successive majorant ratios yield

$$0 < \Phi(u) < 4\pi^2 e^{9u - \pi e^{4u}}. \tag{1.7}$$

Indeed the ratio of successive terms of $n^4 e^{-\pi n^2 e^{4u}}$ is at most $16e^{-3\pi} < 1/2$. The bound justifies differentiation of every order on compact sets of complex (t, z). Thus H is jointly entire, even in z, and real on real (t, z); (1.6) follows under the integral. For each fixed real time it is a real entire function of order at most one, as is also immediate from the same source bound by splitting the integral at $u = O(\log(2 + |z|))$.

1.2 Uniform positivity outside the inner region

The leading result concerns the dimensionless parameter $t \log |x|$, which remains nontrivial as time tends to zero while height increases.

Theorem 1.1 (Uniform four-jet asymptotic)**.** *For every $\varepsilon > 0$ there is $X_\varepsilon > 4\pi$ with the following property. Uniformly for*

$$x \geq X_\varepsilon, \qquad 0 < t \leq \tfrac{1}{2}, \qquad \theta := tL \geq 4 + \varepsilon, \qquad L := \log \frac{x}{4\pi}, \tag{1.8}$$

there is an explicitly defined nonzero complex amplitude $Q = Q_t(x)$ such that, with $\omega = L/4$ and $\phi = \arg Q$,

$$\frac{1}{2|Q|} \left(h_0, \frac{h_1}{\omega}, \frac{h_2}{\omega^2}, \frac{h_3}{\omega^3} \right) = (\cos\phi, -\sin\phi, -\cos\phi, \sin\phi) + O_\varepsilon(L^{-1}). \tag{1.9}$$

Consequently

$$h_0^2 + (h_1/\omega)^2 = 4|Q|^2(1 + O_\varepsilon(L^{-1})), \tag{1.10}$$

$$\mathcal{L}_0 = 4|Q|^2\omega^2(1 + O_\varepsilon(L^{-1})), \qquad \mathcal{L}_1 = 4|Q|^2\omega^4(1 + O_\varepsilon(L^{-1})). \tag{1.11}$$

After increasing X_ε, both Laguerre forms are strictly positive everywhere in (1.8), and $F(t, x) \neq 0$ there. Evenness gives the reflected statements at negative heights.

The theorem is a consequence of the effective approximation in [3, Theorem 1.3], proved below with uniform control of the four derivatives. Its height cutoff is not evaluated numerically here. The number four is the sufficient summability threshold in this proof, not an asserted optimal boundary for positivity.

The geometric interpretation is that a common nonzero amplitude controls all four jets, including at phases where individual leading components vanish. In quaternionic and octonionic coordinates, the same quadratic forms become signed geometric observations. Those constructions do not supply positivity by themselves; they identify precisely which part of the theta analysis is needed. A second, complementary result is the validated bounded cover

$$|F(t, x)| > 10^{-16} \quad (0 \leq t \leq 1/2,\ |x| \leq 104), \tag{1.12}$$

together with both strict stationary signs there. Its complete-source proof and certificate structure are given in Section 6.

Real double zeros at positive times are the relevant obstruction: the classical heat-flow inputs imply

$$\mathrm{RH} \quad \Longleftrightarrow \quad F(t,x) \neq 0 \quad \text{for every } t>0 \text{ and } x \in \mathbb{R}. \tag{1.13}$$

Section 3 gives the reduction and the sufficient stationary bounded-loss criterion. Theorem 1.1 and (1.12) leave an inner small-time region and a finite transition strip. The exact shared-source formula later in the paper formulates the remaining arithmetic estimate without discarding its signed cross terms.

2 Proof of the uniform outer-region theorem

2.1 A holomorphic comparison amplitude

Use the principal logarithm and set

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}\exp\left\{\left(\frac{s}{2}-\frac{1}{2}\right)\operatorname{Log}\frac{s}{2}-\frac{s}{2}\right\}, \tag{2.1}$$

$$\alpha(s) = \frac{M_0'(s)}{M_0(s)} = \frac{1}{2s}+\frac{1}{s-1}+\frac{1}{2}\operatorname{Log}\frac{s}{2\pi}, \qquad M_t(s) = e^{t\alpha(s)^2/4}M_0(s), \tag{2.2}$$

$$s_\pm(z) = \frac{1\pm iz}{2}, \qquad w_\pm(z) = s_\pm(z)+\frac{t}{2}\alpha(s_\pm(z)), \tag{2.3}$$

$$Q_\pm(z) = M_t(s_\pm(z))\zeta(w_\pm(z)), \qquad \mathcal{P}_t(z) = Q_+(z)+Q_-(z). \tag{2.4}$$

For real x we take $Q = Q_+(x)$; the other branch is its conjugate. The functions in these definitions are analytic on the small discs used below. Their arguments stay at imaginary parts approximately $\pm x/2$, away from all logarithmic cuts, while $\Re w_\pm > 1$ keeps the zeta arguments away from the pole.

Write $z = u+iv$, $L_u = \log(u/(4\pi))$, and

$$N = \left\lfloor \sqrt{u/(4\pi)+t/16} \right\rfloor, \qquad S_{t,N}(w) = \sum_{n\le N} e^{t\log^2 n/4} n^{-w}.$$

Polymath's estimate applies when $u \ge 200$, $0 \le v \le 1$, and $0 < t \le 1/2$. In the present notation it reads

$$H(t,z) = M_t(s_-(z))S_{t,N}(w_-(z)) + M_t(s_+(z))S_{t,N}(w_+(z)) + \mathcal{E}(t,z), \tag{2.5}$$

where $|\mathcal{E}| \le |M_t(s_-)|(e_A+e_B+e_{C,0})$. This is the exact two-branch expression of the cited theorem: its first exponent s_* equals w_- and its second exponent, after incorporating the factor n^v, is $\overline{s_*}+\kappa-v = w_+$. To retain the full imported error information, define

$$\gamma = \frac{M_t(s_+)}{M_t(s_-)}, \qquad \kappa = \frac{t}{2}\left(\alpha\left(\frac{1-v+iu}{2}\right)-\alpha\left(\frac{1+v+iu}{2}\right)\right).$$

The bounds needed from [3, eqs. (20)–(24)] are

$$|\gamma| \le e^{0.02v}(u/(4\pi))^{-v/2}, \qquad |\kappa| \le \frac{tv}{2(u-6)}, \tag{2.6}$$

$$e_A+e_B \le \sum_{n\le N}(1+|\gamma|N^{|\kappa|}n^v)e^{t\log^2 n/4}n^{-\Re w_-}\left[\exp\left\{\frac{(t^2/16)\log^2(u/(4\pi n^2))+0.626}{u-6.66}\right\}-1\right], \tag{2.7}$$

$$e_{C,0} \le (u/(4\pi))^{-(1+v)/4} \exp\left\{ -\frac{tL_u^2}{16} + \frac{1.24(3^v + 3^{-v})}{N - 0.125} + \frac{3|L_u + i\pi/2| + 10.44}{u - 12} \right\}. \tag{2.8}$$

No derivatives of this finite approximation will be taken. Its cutoff can jump with u; the analytic model (2.4) will remove that difficulty before differentiation.

2.2 Uniform replacement of the weighted sums

Fix (t, x) in (1.8) and put

$$D_x = \{z : |z - x| \le L^{-1}\}, \qquad \theta_u = tL_u. \tag{2.9}$$

Throughout this disc, $L_u = L + O((xL)^{-1})$, $|v| \le L^{-1}$, and $\theta_u \ge 4 + \varepsilon/2$ for sufficiently large x depending on ε. All constants below are independent of t in the stated range. The elementary formula for α gives

$$\Re w_\pm \ge \frac{1}{2} - \frac{|v|}{2} + \frac{\theta_u}{4} - O(t/u), \qquad \log N \le \frac{L_u}{2} + O(u^{-1}). \tag{2.10}$$

For $n \le N$ it follows that

$$e^{t \log^2 n/4} n^{-\Re w_\pm} \le n^{-p}, \qquad p = \frac{1}{2} + \frac{\theta_u}{8} - \delta_x, \quad 0 \le \delta_x = \frac{|v|}{2} + O(u^{-1}). \tag{2.11}$$

Choose x large enough that $\delta_x \le \varepsilon/32$. With $\theta_0 = 4 + \varepsilon/2$, this gives

$$p \ge 1 + \frac{\varepsilon}{32} + \frac{\theta_u - \theta_0}{8}. \tag{2.12}$$

The weighted absolute sums are uniformly bounded, and (2.10) also gives the convenient weaker bound $\Re w_\pm \ge 5/4$.

Using $e^q - 1 \le qe^q$ for $q \ge 0$, we have

$$|S_{t,N}(w) - \zeta(w)| \le \frac{t}{4} \sum_{n=2}^{\infty} (\log n)^2 n^{-p} + \frac{N^{1-\Re w}}{\Re w - 1}. \tag{2.13}$$

The dependence on time needs care: θ_u may grow with x, so a fixed-θ argument would not suffice. Put $r = \theta_u - \theta_0 \ge 0$ and $D_\varepsilon = \sum_{n \ge 2} (\log n)^2 n^{-1-\varepsilon/32} < \infty$. Then the first term in (2.13) is at most

$$\frac{D_\varepsilon}{4L_u} (\theta_0 + r) 2^{-r/8} = O_\varepsilon(L^{-1}), \tag{2.14}$$

because the numerator is bounded for $r \ge 0$. The ordinary zeta tail is $O(N^{-1/4}) = O(x^{-1/8})$. Hence, uniformly on the whole disc,

$$S_{t,N}(w_\pm) - \zeta(w_\pm) = O_\varepsilon(L^{-1}). \tag{2.15}$$

The threshold four enters precisely through the strict inequality $1/2 + \theta/8 > 1$ in the weighted absolute sum.

2.3 Complete value errors and derivative control

On the upper half of D_x, (2.6) bounds the extra factors in (2.7). Multiplication by n^v changes the leading real part $(1+v)/2$ to $(1-v)/2$, already covered by (2.11). Also $|\log(u/(4\pi n^2))| \le L_u + O(u^{-1})$ for $n \le N$. Since $\theta_u \le L_u/2$, the exponential error multiplier in (2.7) is $O((1+\theta_u^2)/u)$. Equations (2.7)–(2.8) therefore give

$$e_A + e_B = O_\varepsilon(L^2/x), \qquad e_{C,0} = O_\varepsilon(x^{-1/2}). \tag{2.16}$$

For the latter bound use $\exp(-L_u/4 - \theta_u L_u/16) \le C_\varepsilon x^{-1/2}$; the remaining correction exponent in (2.8) tends uniformly to zero.

Let $\Gamma = |M_t(s_+(x))|$. The logarithmic z-derivative of either $M_t(s_\pm(z))$ is $O(L)$ uniformly for $0 < t \le 1/2$. Integrating along paths of length at most $1/L$ yields

$$|M_t(s_\pm(z))| \asymp \Gamma \qquad (z \in D_x). \tag{2.17}$$

This is the reason for the shrinking disc: a fixed-radius disc would permit powers of x in this comparison. Combining (2.5), (2.15), (2.16), and (2.17) gives

$$H(t,z) - \mathcal{P}_t(z) = O_\varepsilon(\Gamma/L). \tag{2.18}$$

Initially this holds on the upper half-disc. Schwarz symmetry of both H and $\mathcal{P}_t$ extends it to the lower half, without any new hypothesis on the imported approximation.

Absolute convergence of the Dirichlet series for ζ and $1/\zeta$ in $\Re w \ge 5/4$ gives

$$\zeta(5/4)^{-1} \le |\zeta(w)| \le \zeta(5/4). \tag{2.19}$$

Thus $Q \ne 0$ and $|Q| \asymp \Gamma$. This lower bound is obtained in a known zero-free half-plane and does not assume a physical gap for H or its derivatives. The difference in (2.18) is holomorphic on a neighborhood of the closed disc. Cauchy's formula, with time fixed, now gives

$$\partial_x^k(H(t,\cdot) - \mathcal{P}_t)(x) = O_{\varepsilon,k}(|Q|L^{k-1}), \qquad k = 0,1,2,3. \tag{2.20}$$

The $k = 0$ error is $O_\varepsilon(|Q|/L)$. The pointwise finite-sum bound has been replaced by a holomorphic comparison before applying Cauchy's formula, so no moving-cutoff derivative is present.

2.4 The common phase and the two signs

At the real center,

$$\alpha(s_+(x)) = L/2 + i\pi/4 + O(x^{-1}), \qquad \alpha^{(r)}(s_+(x)) = O_r(x^{-r}) \quad (r \ge 1). \tag{2.21}$$

All fixed logarithmic derivatives of ζ are uniformly bounded in $\Re w \ge 5/4$. Differentiating a local logarithm of Q_+ gives

$$(\log Q_+)' = \frac{i}{2}\left[\alpha + \frac{t}{2}\alpha\alpha' + \left(1 + \frac{t}{2}\alpha'\right)\frac{\zeta'(w_+)}{\zeta(w_+)}\right] = i\omega + O(1), \tag{2.22}$$

$$(\log Q_+)'' = O(1), \qquad (\log Q_+)''' = O(1). \tag{2.23}$$

Here all α terms are evaluated at $s_+(x)$, and time is held fixed. The bounds remain uniform as $\theta = tL$ grows because $t \le 1/2$. The exponential derivative formulas imply

$$Q_+^{(k)}(x) = Q((i\omega)^k + O(L^{k-1})), \qquad k = 1,2,3.$$

The derivatives of Q_- are their conjugates on the real axis. Together with (2.20), this proves (1.9).

To pass to signs, set $P_k = h_k/(2|Q|\omega^k)$ and write $P = (c, -s, -c, s) + e$, where $c^2 + s^2 = 1$ and every component of e is $O_\varepsilon(L^{-1})$. Direct expansion yields

$$P_0^2 + P_1^2 = 1 + O_\varepsilon(L^{-1}), \quad P_1^2 - P_0P_2 = 1 + O_\varepsilon(L^{-1}), \quad P_2^2 - P_1P_3 = 1 + O_\varepsilon(L^{-1}).$$

This proves (1.10)–(1.11), uniformly in phase, and finishes the proof of Theorem 1.1.

There is an explicit algebraic stopping condition for any future numerical evaluation of X_ε. If $|e_j| \le \delta$, then

$$P_0^2 + P_1^2 \ge 1 - 4\delta, \qquad P_1^2 - P_0P_2,\ P_2^2 - P_1P_3 \ge 1 - 4\delta - \delta^2. \tag{2.24}$$

For example the first signed form is $1 - 2se_1 + e_1^2 - ce_2 + ce_0 - e_0e_2$; keep its nonnegative square and bound the other terms. At $\delta = 1/8$ the two signed margins are at least $31/64$ and the balanced norm is at least $1/2$.

Corollary 2.1 (Asymptotic collision localization)**.** *If $H(T_m, a_m) = H_x(T_m, a_m) = 0$, $0 < T_m \le 1/2$, and $|a_m| \to \infty$, then*

$$\limsup_{m\to\infty} T_m \log \frac{|a_m|}{4\pi} \le 4.$$

Proof. For each fixed $\varepsilon > 0$, Theorem 1.1 excludes all sufficiently high terms with the displayed product at least $4 + \varepsilon$. □

The corollary is asymptotic. It does not replace the localization constant by exactly four at every finite height, and it gives no uniform conclusion on the boundary obtained by setting $\varepsilon = 0$.

3 Why positive-time collisions are the obstruction

The preceding asymptotic result is useful because a failure of RH forces a finite collision of real zeros at a positive time. We record the precise reduction, including the compactness input, before formulating the signed estimate still needed outside the proved regions.

We use the normalization

$$H(0, z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right), \qquad H_t = -H_{zz}.$$

The positive, rapidly decreasing theta source makes H jointly entire in (t, z), even in z, and real on the real (t, z) plane. Moreover, $H(t, iy) > 0$ for real t, y, so no time slice is identically zero and every zero has finite multiplicity.

The classical de Bruijn–Newman theory gives a finite real threshold Λ such that all zeros of $H(t, \cdot)$ are real exactly when $t \ge \Lambda$; the endpoint is included [1, 2]. In this normalization $\Lambda \le 1/2$ and RH is equivalent to $\Lambda \le 0$. De Bruijn's strip contraction keeps all zeros in $|\operatorname{Im} z| \le 1$ for $t \ge 0$. We also use the uniform confinement theorem of [3, Theorem 1.5(i)]: there is an absolute constant $C > 0$ such that, for $0 < t \le 1/2$, a nonreal zero satisfies

$$|\operatorname{Re} z| < e^{C/t}. \tag{3.1}$$

Uniformity in time, rather than a separate uncontrolled radius for each slice, is what prevents a first collision from escaping to infinity.

Lemma 3.1 (Local splitting at a multiple zero)**.** *Let W be real analytic near (T, a), solve $W_t = -W_{xx}$, and have an exact spatial zero of multiplicity $m \geq 2$ there. Its nearby zeros have leading locations*

$$a + \sqrt{2(t-T)}\,\lambda_k + O(|t-T|), \qquad k = 1, \dots, m,$$

where the λ_k are the distinct real roots of the probabilists' Hermite polynomial He_m*. In particular, W has a nonreal zero at times immediately before T.*

Proof. Put $W_q = \partial_x^q W(T, a)$. The convergent heat Taylor series is

$$W(T+\delta, a+w) = \sum_{p,q\geq 0} \frac{(-\delta)^p w^q}{p!q!} W_{2p+q}.$$

With $\delta = \eta^2$ and $w = \sqrt{2}\eta\lambda$, division by η^m extends analytically to $\eta = 0$, where its value is $2^{m/2} W_m \,\mathrm{He}_m(\lambda)/m!$. The roots of He_m are real and simple: its Rodrigues formula gives orthogonality to all lower-degree polynomials for the positive Gaussian weight, and the polynomial formed from its real sign-changing factors would contradict that orthogonality unless there were m such roots. The simple roots therefore continue by the implicit function theorem. For $t < T$, take $\eta = i\sqrt{T-t}$; at least one Hermite root is nonzero, so its leading imaginary displacement cannot be cancelled by the $O(T-t)$ remainder. □

Proposition 3.2 (Finite collision criterion for RH)**.** *RH holds if and only if*

$$H(t,x) = H_x(t,x) = 0 \quad \text{has no solution with } 0 < t \leq 1/2,\ x \in \mathbb{R}. \tag{3.2}$$

Equivalently, $E(t,x) = h_0(t,x)^2 + h_1(t,x)^2$ is positive throughout this region. No uniform positive lower bound for E is required. Equivalently, there is no such joint zero at any positive time.

Proof. If RH holds, all zeros are real at all nonnegative times. A multiple zero at a positive time would contradict Lemma 3.1.

Conversely, if RH fails then $0 < \Lambda \leq 1/2$. Choose $t_k \uparrow \Lambda$ with $t_k \geq \Lambda/2$, and a nonreal zero z_k at each t_k. The strip bound and (3.1) put these zeros in the fixed compact set $|\operatorname{Im} z| \leq 1$, $|\operatorname{Re} z| \leq e^{2C/\Lambda}$. A subsequence converges to a zero a of $H(\Lambda, \cdot)$, which is real by the threshold theorem. If that zero were simple, the local implicit zero branch would be unique; conjugation symmetry would make it real for real times on both sides of Λ. This contradicts $z_k \notin \mathbb{R}$. Thus $H(\Lambda, a) = H_x(\Lambda, a) = 0$. The source positivity gives $H(\Lambda, 0) > 0$, so $a \neq 0$. □

3.1 A sufficient estimate only on two stationary sets

Recall

$$\mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2}, \qquad j = 0, 1.$$

On a smooth stationary curve $h_{j+1}(t, \chi(t)) = 0$, the heat equation gives

$$\frac{1}{2}\frac{d}{dt} h_j(t, \chi(t))^2 = \mathcal{L}_j(t, \chi(t)).$$

The following result includes singular stationary sets and all finite multiplicities. Its losses may grow with height or as $t \downarrow 0$; only local boundedness through each possible positive-time collision is required.

Lemma 3.3 (A stationary predecessor of an even zero)**.** *Suppose $W_t = -W_{xx}$ and W has exact spatial multiplicity $2r$ at (T, a), where $r \geq 1$. With $W_q = \partial_x^q W(T, a)$, there is a real analytic function $\chi(h)$ for small $h \geq 0$ such that*

$$W_x(T-h, \chi(h)) = 0, \qquad \chi(h) = a - \frac{W_{2r+1}}{rW_{2r}}h + O(h^2),$$
$$W(T-h, \chi(h)) = \frac{W_{2r}}{r!}h^r + O(h^{r+1}), \qquad W_{xx}(T-h, \chi(h)) = \frac{W_{2r}}{(r-1)!}h^{r-1} + O(h^r).$$

Proof. Use the heat Taylor series above with $\delta = -h$ and $w = hy$. The function $h^{-r}W_x(T-h, a+hy)$ extends analytically to $h = 0$, where it equals $W_{2r}y/(r-1)! + W_{2r+1}/r!$. Its y derivative is nonzero. The implicit function theorem gives $y(h)$, and substitution in the same series gives the two displayed asymptotics. □

Theorem 3.4 (Stationary bounded-loss criterion)**.** *Suppose there are nonnegative functions K_0, K_1, locally bounded on $(0, 1/2] \times \mathbb{R}$, for which*

$$\begin{aligned} h_1 = 0 \quad &\Longrightarrow \quad \mathcal{L}_0 \geq -K_0 E, \\ h_2 = 0 \quad &\Longrightarrow \quad \mathcal{L}_1 \geq -K_1 E. \end{aligned} \tag{3.3}$$

Then RH holds. Local boundedness here includes a neighborhood of every candidate joint zero, using a left neighborhood in time at $t = 1/2$.

Proof. Suppose H has a real multiple zero at (T, a) in this time range, of exact multiplicity $m \geq 2$. If $m = 2r$, apply Lemma 3.3 to $W = H$. On its predecessor branch $h_1 = 0$, $E = H^2 > 0$, and

$$-\frac{\mathcal{L}_0}{E} = \frac{H_{xx}}{H} = \frac{r}{h} + O(1), \qquad h = T - t \downarrow 0.$$

This contradicts the first inequality with locally bounded K_0. If $m = 2r + 1$, apply the lemma to $W = H_x$. Then $h_2 = 0$, $H_x = ch^r + O(h^{r+1})$, and $H_{xxx} = rch^{r-1} + O(h^r)$, with $c \neq 0$. The heat Taylor series also gives $H = O(h^{r+1})$ on this branch, because its displacement from a is $O(h)$. Consequently

$$E = H_x^2(1 + O(h^2)), \qquad -\frac{\mathcal{L}_1}{E} = \frac{r}{h} + O(1),$$

contradicting the second inequality. All divisions occur on proved nonzero predecessor branches, not at the supposed collision. Proposition 3.2 now applies. □

Both levels matter: the heat polynomial $H(t, x) = x^3 - 6(t - T)x$ has $\mathcal{L}_0 = 3x^4 + 36(t - T)^2 \geq 0$ everywhere, but has a triple zero at $(T, 0)$. Its second stationary loss diverges on approaching that collision from earlier times. Thus positivity of the first observation alone is insufficient for arbitrary multiplicity.

4 The four jets as quaternionic and octonionic fields

The four scalar jets enter the collision problem in overlapping pairs: H, H_x detect a multiple zero, while the two stationary signs involve (H, H_x, H_{xx}) and (H_x, H_{xx}, H_{xxx}). Quaternion multiplication keeps both the norms of these pairs and their signed determinants. A compatible octonionic extension retains these observations as oriented associators. We give the constructions explicitly so that their analytic content and their limitations are visible.

4.1 A quaternionic heat field with independent external time

Use Hamilton units $\mathbf{ij} = \mathbf{k}$, $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$, and identify $\mathbb{C}$ with $\mathbb{C}_\mathbf{i} = \mathbb{R} + \mathbf{i}\mathbb{R}$. Complex coefficients are placed on the right of $\mathbf{j}$. Thus

$$(a + \mathbf{j}b)(c + \mathbf{j}d) = ac - \bar{b}d + \mathbf{j}(bc + \bar{a}d), \qquad \overline{a + \mathbf{j}b} = \bar{a} - \mathbf{j}b, \tag{4.1}$$

for $a, b, c, d \in \mathbb{C}_\mathbf{i}$, and $|a + \mathbf{j}b|^2 = |a|^2 + |b|^2$.

Write the independent quaternion coordinates as

$$q = x_0 + \mathbf{i}x_1 + \mathbf{j}x_2 + \mathbf{k}x_3 = z_0 + \mathbf{j}z_1, \qquad z_0 = x_0 + \mathbf{i}x_1, \quad z_1 = x_2 - \mathbf{i}x_3.$$

The unnormalized left Fueter operator and its conjugate are

$$D_q = \partial_{x_0} + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}, \qquad \overline{D}_q = \partial_{x_0} - \mathbf{i}\partial_{x_1} - \mathbf{j}\partial_{x_2} - \mathbf{k}\partial_{x_3}.$$

A differentiable field is *left Fueter regular* if $D_q F = 0$. The convention using the factor $1/2$ has the same kernel; all units here multiply on the left. For the classical theory see [4].

Let $H(\tau, z)$ be the jointly entire theta heat family, satisfying $H_\tau = -H_{zz}$. With $t \in \mathbb{R}$ an independent external parameter, set

$$\tau = t + \overline{z_1} = t + x_2 + \mathbf{i}x_3.$$

$$G_m(t, q) = \partial_z^m H(\tau, z_0) + \mathbf{j}\partial_z^{m+1} H(\tau, z_0), \qquad F = G_0. \tag{4.2}$$

The coordinate z_1 supplies a complex heat-time displacement; both entries of G_m have the same spectral argument.

Proposition 4.1 (Regular lift and jet identities)**.** *Each G_m is globally real analytic and left Fueter regular in q. For every $m \geq 0$,*

$$D_q G_m = 0, \qquad \overline{D}_q G_m = 2G_{m+1}, \qquad \partial_t G_m = -G_{m+2}. \tag{4.3}$$

$$\begin{aligned} \partial_{x_0} G_m &= G_{m+1}, & \partial_{x_1} G_m &= G_{m+1}\mathbf{i}, \\ \partial_{x_2} G_m &= -G_{m+2}, & \partial_{x_3} G_m &= -G_{m+2}\mathbf{i}. \end{aligned} \tag{4.4}$$

Proof. For complex-valued U, V, direct use of (4.1) gives

$$\tfrac{1}{2} D_q(U + \mathbf{j}V) = U_{\bar{z}_0} - V_{z_1} + \mathbf{j}(V_{z_0} + U_{\bar{z}_1}).$$

The derivatives are componentwise Wirtinger derivatives. Taking $U = \partial_z^m H(\tau, z_0)$ and $V = \partial_z^{m+1} H(\tau, z_0)$, the first two terms vanish by holomorphy and the final pair cancel by the heat equation. Joint entireness proves global real analyticity. Differentiating the declared arguments proves (4.4) and the external time identity. Finally $D_q + \overline{D}_q = 2\partial_{x_0}$ proves the remaining assertion. □

On the physical slice $q = x \in \mathbb{R}$, write $h_m = h_m(t, x) = \partial_x^m H(t, x)$. The jets are real and

$$F = h_0 + \mathbf{j}h_1, \qquad F_x = G_1 = h_1 + \mathbf{j}h_2, \qquad F_{xx} = G_2 = h_2 + \mathbf{j}h_3 = -F_t. \tag{4.5}$$

Thus the four physical jets occur in three consecutive overlapping quaternion pairs. The basic observations are

$$E = |F|^2 = h_0^2 + h_1^2, \qquad \mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2} \quad (j = 0, 1). \tag{4.6}$$

In particular $F(t,x) = 0$ means exactly that $H(t,\cdot)$ has a multiple real zero at x. The scalar positive-time collision criterion therefore translates to

$$\text{RH} \quad \Longleftrightarrow \quad F(t,x) \neq 0 \quad (t > 0,\ x \in \mathbb{R}).$$

This uses the scalar collision theorem for the theta family. It makes no assertion about zeros at nonreal quaternion arguments.

The signed products and the heat equation give, on this slice,

$$\overline{G_j} G_{j+1} = h_j h_{j+1} + h_{j+1} h_{j+2} - \mathbf{j}\mathcal{L}_j, \tag{4.7}$$

$$E_t = -2(h_0 h_2 + h_1 h_3), \qquad (\partial_t + \partial_x^2) E = 2(h_1^2 + h_2^2). \tag{4.8}$$

Consequently

$$h_1 = 0 \implies E_t = 2\mathcal{L}_0, \qquad h_2 = 0 \implies E_t = 2\mathcal{L}_1. \tag{4.9}$$

These formulas preserve the two different stationary conditions. The positive source term in the backward-heat identity (4.8) does not give a forward minimum principle.

4.2 An eight-dimensional monogenic extension

For a reader unfamiliar with octonions, it suffices here to regard $\mathbb{O}$ as two copies of $\mathbb{H}$, with a specified multiplication. Write $(a,b) = a + \ell b$, where $\ell = (0,1)$, and adopt the Cayley–Dickson convention

$$(a,b)(c,d) = (ac - d\bar{b}, \bar{a}d + cb), \qquad \overline{(a,b)} = (\bar{a}, -b). \tag{4.10}$$

Then $|(a,b)|^2 = |a|^2 + |b|^2$. This is the convention in [5, Section 2.2]. Octonionic multiplication is nonassociative; we keep parentheses whenever three octonionic factors occur. All component products on the right of (4.10) are associative quaternion products.

Let $r = y_0 + \mathbf{i}y_1 + \mathbf{j}y_2 + \mathbf{k}y_3$, so that $p = q + \ell r$ is the eight-dimensional argument. Put $e_0 = 1, e_1 = \mathbf{i}, e_2 = \mathbf{j}, e_3 = \mathbf{k}$, and define the left operator

$$\mathcal{D}_8 = \sum_{\mu=0}^{3} e_\mu \partial_{x_\mu} + \sum_{\mu=0}^{3} (\ell e_\mu) \partial_{y_\mu}. \tag{4.11}$$

We call a field *left monogenic* if $\mathcal{D}_8 \mathcal{O} = 0$. This fixes the operator convention explicitly; general octonionic Cauchy–Riemann systems are discussed in [6].

Proposition 4.2 (Explicit octonionic extension)**.** *For each $m \geq 0$, the field*

$$\mathcal{O}_m(t,q,r) = \Big(G_m(t,q) - \tfrac{1}{2} G_{m+3}(t,q)\bar{r},\, G_{m+2}(t,q) \Big) \tag{4.12}$$

is globally real analytic and satisfies

$$\mathcal{D}_8 \mathcal{O}_m = 0, \qquad \partial_{x_0} \mathcal{O}_m = \mathcal{O}_{m+1}, \qquad \partial_t \mathcal{O}_m = -\mathcal{O}_{m+2}, \qquad \mathcal{O}_m(t,q,0) = (G_m, G_{m+2}). \tag{4.13}$$

Proof. Applying (4.10) separately to each term of (4.11) gives the component formula

$$\mathcal{D}_8(A,B) = \left(D_q A - \sum_{\mu=0}^{3} B_{y_\mu} \bar{e}_\mu,\, \overline{D}_q B + \sum_{\mu=0}^{3} A_{y_\mu} e_\mu \right). \tag{4.14}$$

For (4.12), B is independent of r. Moreover $D_q(G_{m+3}\overline{r}) = (D_qG_{m+3})\overline{r} = 0$, since the component algebra is associative and r is held fixed. The first component of (4.14) is therefore zero. The second is

$$2G_{m+3} - \tfrac{1}{2}\sum_{\mu=0}^{3} G_{m+3}\overline{e}_\mu e_\mu = 2G_{m+3} - 2G_{m+3} = 0.$$

The derivative identities follow directly from (4.3), and the explicit formula proves global real analyticity. □

The correction is necessary: the extension (G_m, G_{m+2}) constant in r has defect $(0, 2G_{m+3})$. More generally, two left Fueter-regular fields $A(q), B(q)$ admit the extension $(A - \frac{1}{2}B_{x_0}\overline{r}, B)$, by the same calculation. Thus the construction imposes no new condition on the original pair. For $m = 0$, its value away from $r = 0$ involves G_3, hence h_4; it is not a closed evolution of only four jets.

On the physical trace $(q, r) = (x, 0)$,

$$\mathcal{O}_0 = (h_0 + \mathbf{j}h_1) + \ell(h_2 + \mathbf{j}h_3) \in \mathrm{span}_{\mathbb{R}}\{1, \mathbf{j}, \ell, \ell\mathbf{j}\}.$$

This span is an associative quaternion subalgebra, as can be checked from (4.10). Its total norm is

$$|\mathcal{O}_0|^2 = E + h_2^2 + h_3^2. \tag{4.15}$$

At a nondegenerate double zero, $E = 0$ but $h_2 \neq 0$; the total octonionic norm is then positive. A collision criterion must retain the distinguished first pair.

4.3 Oriented associators preserve the signed observations

The *associator* measures the failure of associativity:

$$[A, B, C] = (AB)C - A(BC), \qquad \mathbf{u} = \ell\mathbf{j}.$$

For embedded quaternions $a, b \in \mathbb{H}$, the pair multiplication gives

$$[a, b, \ell] = \ell(\overline{ab} - \overline{ba}) = -\ell(ab - ba). \tag{4.16}$$

The last equality holds because a quaternion commutator has zero real part. The untwisted physical jets lie in the commutative plane $\mathbb{R} + \mathbf{j}\mathbb{R}$, so their associator with ℓ vanishes. Fixed right multiplication by $\mathbf{i}$ and $\mathbf{k}$ produces the needed signed quantities while preserving quaternion left regularity.

Lemma 4.3 (Dot products and determinants as associators)**.** *For $a = A + \mathbf{j}B$, $b = C + \mathbf{j}D$ with real A, B, C, D,*

$$[a\mathbf{i}, b\mathbf{i}, \ell] = -2\mathbf{u}(AD - BC), \qquad [a\mathbf{i}, b\mathbf{k}, \ell] = 2\mathbf{u}(AC + BD). \tag{4.17}$$

Proof. One has $a\mathbf{i} = A\mathbf{i} - B\mathbf{k}$, $b\mathbf{i} = C\mathbf{i} - D\mathbf{k}$, and $b\mathbf{k} = D\mathbf{i} + C\mathbf{k}$. Their quaternion commutators are, respectively, $2\mathbf{j}(AD - BC)$ and $-2\mathbf{j}(AC + BD)$. Substitute these into (4.16). □

In particular define, on the physical slice,

$$\mathcal{A}_E = [F\mathbf{i}, F\mathbf{k}, \ell] = 2\mathbf{u}E, \tag{4.18}$$

$$\mathcal{A}_j = [G_j\mathbf{i}, G_{j+1}\mathbf{i}, \ell] = 2\mathbf{u}\mathcal{L}_j \quad (j = 0, 1), \tag{4.19}$$

$$\mathcal{A}_t = [F\mathbf{i}, F_t\mathbf{k}, \ell] = \mathbf{u}E_t. \tag{4.20}$$

These are immediate from Lemma 4.3 and (4.8). In contrast to (4.15), $\mathcal{A}_E = 0$ is equivalent to $F = 0$. Equation (4.9) becomes

$$h_1 = 0 \implies \mathcal{A}_t = \mathcal{A}_0, \qquad h_2 = 0 \implies \mathcal{A}_t = \mathcal{A}_1.$$

There is no order on $\mathbb{O}$ being invoked. Specify the oriented ray $\mathcal{C}_{\mathbf{u}} = \{s\mathbf{u} : s \geq 0\}$. The two stationary sign conditions are exactly

$$\mathcal{A}_t \in \mathcal{C}_{\mathbf{u}} \quad \text{on} \quad \{h_1 h_2 = 0\}. \tag{4.21}$$

Likewise the locally bounded loss condition is exactly

$$\mathcal{A}_t + K\mathcal{A}_E \in \mathcal{C}_{\mathbf{u}} \quad \text{on} \quad \{h_1 h_2 = 0\}, \qquad K \geq 0, \tag{4.22}$$

with K locally bounded through every candidate collision. Indeed the coefficient on this ray is $E_t + 2KE$. Thus the associators preserve both the collision detector and the required orientation. Their norms alone would replace the signed quantities by absolute values.

4.4 Why a source-specific estimate remains necessary

The constructions above apply to every jointly entire backward-heat family real on the physical slice. The following control shows that even strict positivity and analytic decay of its cosine-transform source do not supply the missing sign.

Proposition 4.4 (A positive entire source with a positive-time collision)**.** *There exist strictly positive, even, entire functions Φ_*, decaying superexponentially on the real axis, whose heat cosine transforms are jointly entire and have a nondegenerate real double zero at a positive time. Their quaternionic and octonionic lifts satisfy all the preceding identities.*

Proof. Set $\rho(v) = 2e^{-v^4}/\Gamma(1/4)$, so that $\int_{\mathbb{R}} \rho(v)\,dv = 1$, and $\rho_\epsilon(v) = \epsilon^{-1}\rho(v/\epsilon)$ for real $\epsilon > 0$. Use

$$\Phi_\epsilon(u) = 2\{\rho_\epsilon(u-1) + \rho_\epsilon(u+1)\} + \rho_\epsilon(u-2) + \rho_\epsilon(u+2).$$

This source is strictly positive, even, and entire, with quartic decay on the real axis. Its half-line heat cosine transform is

$$\begin{aligned}\widetilde{H}_\epsilon(t,z) = 2\int_{\mathbb{R}} \rho(v) e^{t(1+\epsilon v)^2} \cos\bigl(z(1+\epsilon v)\bigr)\,dv \\ + \int_{\mathbb{R}} \rho(v) e^{t(2+\epsilon v)^2} \cos\bigl(z(2+\epsilon v)\bigr)\,dv.\end{aligned}$$

Quartic decay supplies an integrable majorant on every compact set of complex (t, z, ϵ). Hence this formula extends jointly entirely through $\epsilon = 0$, obeys backward heat, and there becomes

$$\widetilde{H}_0(t,z) = 2e^t \cos z + e^{4t}\cos(2z).$$

At $T_0 = (\log 2)/3$, $x_0 = \pi$, it satisfies $\widetilde{H}_0 = (\widetilde{H}_0)_x = 0$ and $(\widetilde{H}_0)_{xx} = -6e^{T_0} \neq 0$. By the heat equation the Jacobian of $(\widetilde{H}_\epsilon, (\widetilde{H}_\epsilon)_x)$ with respect to (t, x), at this point and $\epsilon = 0$, has determinant $-((\widetilde{H}_0)_{xx})^2 \neq 0$. The real analytic implicit-function theorem therefore gives a nearby nondegenerate real double zero at a positive time for every sufficiently small $\epsilon > 0$. □

The failure is visible in the signed observation. At such a collision (T, a), put $c = H_{xx}(T, a) \neq 0$. The implicit-function theorem gives a real stationary branch $H_x(t, \chi(t)) = 0$ through it. For $\delta = T - t \downarrow 0$, backward heat gives

$$H(t, \chi(t)) = c\delta + O(\delta^2), \qquad E(t, \chi(t)) = c^2\delta^2 + O(\delta^3), \qquad \mathcal{L}_0(t, \chi(t)) = -c^2\delta + O(\delta^2).$$

Thus (4.22) would require $K \geq \delta^{-1} + O(1)$, contradicting local boundedness at the collision. The lift and associator identities alone cannot provide that bound. For the theta family, its particular arithmetic structure must enter a further signed estimate.

For the genuine theta family, Theorem 1.1 supplies the additional sign information in its declared outer region: $\mathcal{A}_0$ and $\mathcal{A}_1$ lie strictly on $\mathcal{C}_{\mathbf{u}}$ there. At genuine stationary points the same is true of $\mathcal{A}_t$. This is the geometric consequence of the common arithmetic amplitude, not a consequence of monogenicity alone.

5 Centered residuals and the common theta carrier

The outer theorem uses an absolutely convergent arithmetic comparison. To study the remaining region, we return to the exact theta source. The aim is to retain the same four derivatives and their signed correlations before estimating any error.

5.1 A contour identity for the physical derivatives

Proposition 5.1 (Complete-source contour shift)**.** *For real $t \geq 0$, $x \geq 0$, $0 < a < \pi/8$, and every integer $k \geq 0$,*

$$h_k(t, x) = e^{-ax}\Re \int_0^\infty (i(u + ia))^k e^{t(u+ia)^2} \Phi(u + ia) e^{ixu}\, du. \tag{5.1}$$

Proof. The theta series is normally convergent in the strip $|\Im z| < \pi/8$ because $\Re(e^{4z}) > 0$ there. Analytic continuation of the real reflection identity makes Φ even throughout the strip; Schwarz symmetry then makes $\Phi(iy)$ real. On the left edge of the contour shift, every factor of $(iz)^k e^{tz^2} \Phi(z) e^{ixz}$ is real at $z = iy$, while $dz = i\, dy$. Its contribution has zero real part. On the right edge, $\cos(4y) \geq \cos(4a) > 0$ gives uniform double-exponential decay. Cauchy's theorem proves the formula. The evenness argument applies to the complete source; it must precede any separation into individual theta summands. □

The all-index and integration bounds in Appendix A make truncation of (5.1) rigorous. They control absolute errors at the gamma exponential scale. Obtaining a sign still requires information about the retained, correlated jets.

5.2 A residual adapted to genuine stationarity

Freeze real a, b with $0 < a < \pi/8$ and $b \geq 0$, and put $\mu = a^2 + b^2$. For $j = 0, 1$, define

$$C_j = e^{-ax}\Re \int_0^\infty (u^2 - b^2)(i(u + ia))^j e^{t(u+ia)^2} \Phi(u + ia) e^{ixu}\, du. \tag{5.2}$$

The polynomial identity $u^2 - b^2 = -(i(u + ia))^2 - 2a\, i(u + ia) - \mu$ gives

$$C_j = -h_{j+2} - 2ah_{j+1} - \mu h_j, \qquad h_{j+1} = 0 \implies \mathcal{L}_j = \mu h_j^2 + h_j C_j. \tag{5.3}$$

Thus the contour drift disappears at exactly the original stationary constraint. No derivative of an exponentially normalized observation has replaced that constraint.

Proposition 5.2 (A sufficient residual cone)**.** *If $\eta > 0$ and*

$$\Delta := \eta^2 E - C_0^2 - C_1^2 \geq 0, \tag{5.4}$$

then the respective stationary sets satisfy

$$\mathcal{L}_0 \geq (\mu - \eta)E \quad (h_1 = 0), \qquad \mathcal{L}_1 \geq -\frac{\eta^2}{4\mu}E \quad (h_2 = 0). \tag{5.5}$$

A sufficient common negative-loss coefficient is $K = \eta^2/(4\mu)$. It is locally bounded wherever the prescribed parameters are locally bounded and μ stays positive.

Proof. At $h_1 = 0$, $E = h_0^2$ and $|C_0| \leq \eta\sqrt{E}$, so the first inequality follows from (5.3). At $h_2 = 0$, complete the square in $\mu h_1^2 - \eta|h_1|\sqrt{E}$ to obtain the second. Finally $\eta - \mu \leq \eta^2/(4\mu)$ because $(\eta - 2\mu)^2 \geq 0$. No division by E occurs. □

For use with exact algebra or interval certificates, the same argument has the following identities before stationarity is imposed:

$$\begin{aligned} 2\eta[\mathcal{L}_0 + (\eta - \mu)E] &= (C_0 + \eta h_0)^2 + C_1^2 + \Delta \\ &\quad + 2\eta h_1[(1 - \mu + \eta/2)h_1 + 2ah_0], \end{aligned} \tag{5.6}$$

$$4\mu\mathcal{L}_1 + \eta^2 E = (2\mu h_1 + C_1)^2 + C_0^2 + \Delta + 4\mu h_2(h_2 + 2ah_1). \tag{5.7}$$

Each final term is a multiple of the genuine stationary derivative. The cone is sufficient, but a one-sided bound for $h_j C_j$ can be substantially weaker and therefore more useful.

For example, suppose paid normalized approximations give $Z_j = e^{\pi x/8}h_j = p + \delta$ and $e^{\pi x/8}C_j = q + \epsilon'$, where $|\delta| \leq \varepsilon_0$, $|\epsilon'| \leq \zeta_0$. At $h_{j+1} = 0$, expansion of (5.3), retaining $\mu\delta^2 \geq 0$, gives

$$e^{\pi x/4}\mathcal{L}_j \geq \mu p^2 + pq - |2\mu p + q|\varepsilon_0 - |p|\zeta_0 - \varepsilon_0\zeta_0. \tag{5.8}$$

Every truncation, quadrature, and parameter error must be included in the two bounds. The retained product pq preserves its sign.

5.3 One carrier for all four derivatives

There is an exact arithmetic reduction behind the centered residual. Put $d_n = \frac{1}{2}\log n$. The source summands obey

$$\phi_n(z) = n^{-1/2}\phi_1(z + d_n). \tag{5.9}$$

For $v \geq d_n$, define

$$A_n(t, x; v) = n^{-1/2}\exp\{td_n^2 - 2td_n(v + ia) - ixd_n\}. \tag{5.10}$$

Changing variables $v = u + d_n$ in the complete shifted source yields

$$h_j = e^{-ax}\Re\int_0^\infty e^{t(v+ia)^2}\phi_1(v + ia)e^{ixv}\sum_{n \leq e^{2v}} A_n(t, x; v)[i(v + ia - d_n)]^j\, dv. \tag{5.11}$$

In particular $j = 0, 1, 2, 3$ uses only the first four logarithmic moments of the same arithmetic weights. The staircase cutoff $n \leq e^{2v}$ is part of the identity. It is not an independently chosen truncation.

To justify the rearrangement, let $0 \le t \le \tau$. Since $0 \le d_n \le v$, we have $|A_n| \le n^{-1/2}$ and $\sum_{n \le e^{2v}} n^{-1/2} \le 2e^v$. Therefore the absolute integrand in (5.11), before e^{-ax}, is bounded by

$$2(2\pi^2 + 3\pi)(v+a)^j e^{\tau v^2 + 10v - \pi \cos(4a) e^{4v}}. \tag{5.12}$$

This is integrable. It also gives a full integration tail, stated in Appendix A; no unretained index is omitted from that bound.

The common oscillatory exponential in the original n-th channel has phase $(x+2ta)u - \pi n^2 \sin(4a) e^{4u}$. Its critical location is

$$b_n = \frac{1}{4} \log \frac{x + 2ta}{4\pi n^2 \sin(4a)} = b_1 - d_n. \tag{5.13}$$

Only nonnegative locations belong to the integration half-line. This observation supplies a centering choice; it does not assert first-channel dominance or a full stationary-phase approximation. Taking $b = b_1$, the multiplier defining C_j becomes

$$(v - d_n)^2 - b^2 = \underbrace{-b \log n + \tfrac{1}{4} \log^2 n}_{\delta_n} + 2(b - d_n)(v - b) + (v - b)^2. \tag{5.14}$$

Thus the remaining signed residual involves logarithmic weights of the same arithmetic sum, rather than unrelated errors for each derivative. At $v = b$ those weights have the transparent form

$$A_n = n^{-1/2 - tb - i(x/2 + ta)} e^{t \log^2 n / 4}. \tag{5.15}$$

The factors A_n are complex and their products retain arithmetic phases. A positive-weight covariance inequality does not apply to them.

For perspective, if $b > 0$ the frozen-parameter balanced quaternion $W = h_0 + \mathbf{j}(h_1 + a h_0)/b$ obeys

$$W_t = (b^2 - a^2 - 2ab\mathbf{j})W + R, \qquad R = C_0 + \mathbf{j}(C_1 - aC_0)/b. \tag{5.16}$$

The skew term rotates W without changing its norm. Equation (5.16) separates the expected oscillatory motion from the residual, whereas (5.3) keeps the precise scalar conditions needed for a sign. If a, b vary with time, the additional term is $\mathbf{j}(a_t \Re W/b - b_t \Im_{\mathbf{j}} W/b)$; all cell computations below freeze them before differentiating.

6 The bounded region: a complete certificate through height 104

The bounded calculation supplies a genuine norm gap, so it excludes collisions directly. Its role is to join the central analytic estimates to the large-height theorem, while leaving the intervening and small-$t \log |x|$ regions explicitly visible.

Theorem 6.1 (Certified closed bounded region)**.** *For the complete theta source,*

$$|F(t,x)| = \sqrt{h_0(t,x)^2 + h_1(t,x)^2} > 10^{-16} \quad (0 \le t \le 1/2,\ |x| \le 104). \tag{6.1}$$

On this rectangle, $h_1 = 0$ implies $\mathcal{L}_0 > 0$, and $h_2 = 0$ implies $\mathcal{L}_1 > 0$. The norm bound combines an analytic central estimate with the validated interval records described below.

6.1 A central estimate with no theta truncation

Proposition 6.2 (Central norm gap)**.** *For $0 \le t \le 1$ and $|x| \le 8$, $|F(t,x)| > 1/120000$. Both stationary signs in Theorem 6.1 hold on this larger time interval.*

Proof. Put $w_t(u) = e^{tu^2}\Phi(u)$ and $M_k(t) = \int_0^\infty u^k w_t(u)\,du$. The complete source obeys $0 < \Phi(u) < 4\pi^2 e^{9u-\pi e^{4u}}$ for $u \ge 0$: after dropping each summand's negative term, the ratio of successive $n^4 e^{-\pi e^{4u} n^2}$ terms is at most $16e^{-3\pi} < 1/2$. The following elementary lower cores and upper tails hold uniformly for $0 \le t \le 1$:

core $[0,d]$	$\Phi(u) > c_d$	tail $u \ge b$	$w_t(u) < A_b e^{-q_b(u-b)}$
$d = 1/10$	$6/25$	$b = 7/25$	$(1/20)e^{-28(u-b)}$
$d = 3/20$	$3/25$	$b = 3/10$	$(1/49)e^{-32(u-b)}$
$d = 1/5$	$1/20$		

(6.2)

For completeness, the lower bounds follow from $\phi_1(u) = e^u g(\pi e^{4u})$ with $g(q) = q(2q-3)e^{-q}$ and $g'(q) = -(2q-1)(q-3)e^{-q}$. On the three cores the respective arguments are less than $33/7, 6, 36/5$. The bounds $e < 11/4$, $e^{2/7} > 9/7$, and $e^{1/5} < 5/4$ give the three stated rational lower bounds for g. For the tails, $\psi_t(u) = 9u + tu^2 - \pi e^{4u}$ is strictly concave. At $b = 7/25$ one has $\psi_t(b) < -69/10$, $\psi_t'(b) < -28$; at $b = 3/10$ one has $\psi_t(b) < -38/5$, $\psi_t'(b) < -32$. Together with $4\pi^2 < 40$, $e^{69/10} > 800$, and $e^{38/5} > 1960$, these prove the full upper tails in (6.2). All comparisons use rational Taylor bounds for the indicated exponentials; in particular they do not truncate the theta source.

Integrating $u^2(1 - 25u^2/2)$ over the first positive core and paying its entire negative tail with $b = 7/25$ gives

$$M_2 - \frac{25}{2}M_4 > \frac{37}{500000} - \frac{5}{8}\frac{3497259}{33614000000} = \frac{2413193}{268912000000} > \frac{1}{120000}. \tag{6.3}$$

Indeed the polynomial is nonnegative up to b and is bounded below on the tail by $-(25/2)u^2(u^2 - b^2)$. Likewise, with $f(u) = u^4(1 - 32u^2/3)$ and $b = 3/10$, the three successive cores and the entire negative tail give

$$M_4 - \frac{32}{3}M_6 > \frac{111799}{33600000000} - \frac{32}{147}\frac{85837761}{6710886400000} = \frac{2092283587}{3853516800000000} > \frac{1}{2000000}. \tag{6.4}$$

The positive term is $(6/25)\int_0^{1/10} f + (3/25)\int_{1/10}^{3/20} f + (1/20)\int_{3/20}^{1/5} f$; the tail integral is $\int_b^\infty u^4(u^2 - b^2)e^{-32(u-b)}\,du$. Thus both displayed differences are exact rational integrations.

The global inequalities $\sin y \ge y - y^3/6$ for $y \ge 0$ and $\cos y \ge 1 - y^2/2$ now imply, for $0 < x \le 8$,

$$-h_1 \ge x(M_2 - x^2 M_4/6) > x/120000, \qquad h_3 \ge x(M_4 - x^2 M_6/6) > x/2000000.$$

For $|x| \le 1$, integrate the first core using $\cos(xu) \ge 199/200$, discard the nonnegative middle interval $[1/10, 7/25]$, and pay the entire remaining tail. This gives

$$H(t,x) > \frac{597}{25000} - \frac{1}{560} = \frac{7733}{350000} > \frac{1}{50}.$$

The value bound covers $|x| \le 1$, and the slope bound covers $1 \le |x| \le 8$, proving the norm gap by evenness. The first stationary set is exactly $x = 0$, where $\mathcal{L}_0 = H(t,0)M_2 > 0$. For $0 < |x| \le 8$, $\mathcal{L}_1 = h_2^2 - h_1 h_3 > x^2/240000000000$; at $x = 0$ one has $h_2 = -M_2 < 0$, so the second stationary set is absent. □

6.2 What the interval certificates enclose

The records use Arb/Acb interval arithmetic to enclose the complete physical jets on closed space–time cells. Each enclosure includes the returned finite-quadrature radius, every omitted theta index, the complete integration tail, rounding, and spatial and temporal Taylor remainders. The older compensated calculation also pays the complete Gaussian transforms and their truncation remainders. Finite compensated channels are transported using the raw heat equation followed by their explicit time-dependent compensation; they are not assigned a homogeneous heat equation.

For a heat-transported sequence Y_r, a cell with half-widths d_t, d_x uses

$$Y_r(t_c+\tau, x_c+\eta) = \sum_{\ell=0}^{q-1}\sum_{m=0}^{p-1} \frac{(-\tau)^\ell \eta^m}{\ell! m!} Y_{r+2\ell+m}(t_c, x_c) + \mathcal{R}_r.$$

If M_k bounds the complete kth derivative on the cell, Taylor's theorem first in time and then in space gives

$$|\mathcal{R}_r| \le \sum_{\ell=0}^{q-1} \frac{d_t^\ell}{\ell!}\frac{d_x^p}{p!} M_{r+2\ell+p} + \frac{d_t^q}{q!} M_{r+2q}. \tag{6.5}$$

The higher bands sum the retained theta coefficients before applying the common Taylor monomials. This preserves cancellation between different indices while adding every positive remainder payment.

Stationary decisions concern genuine derivatives. If $h_k = p_k + r_k$ with $|r_k| \le \varepsilon_k$, then at $h_{j+1} = 0$ one has

$$\mathcal{L}_j \ge p_{j+1}^2 - p_j p_{j+2} - (\varepsilon_{j+1}^2 + |p_j|\varepsilon_{j+2} + |p_{j+2}|\varepsilon_j + \varepsilon_j\varepsilon_{j+2}). \tag{6.6}$$

Here $p_{j+1}^2 \le \varepsilon_{j+1}^2$ follows from genuine stationarity; the square payment has a negative sign. The products retain all cross terms between retained channels. A cell is discharged either by exclusion of the stationary derivative or by a strictly positive paid signed lower bound. Monotone endpoint exclusions, where used, employ the next physical derivative and uniform endpoint signs. No zero of an ordinary derivative of an exponentially rescaled function is substituted.

Independently, if I_0, I_1 enclose the genuine h_0, h_1 throughout a cell, then

$$E \ge \operatorname{dist}(0, I_0)^2 + \operatorname{dist}(0, I_1)^2. \tag{6.7}$$

Exact rational comparisons and downward square-root rounding turn these energy floors into the reported norm floors. No division by an unproved norm is used.

6.3 Coverage, the centered-residual pilot, and reproducibility

Every numerical band covers the full closed time interval $[0, 1/2]$ using eight slabs $[r/16, (r+1)/16]$. The records give the following conservative norm bounds; the central row is analytic.

Positive-height band	Cells	Precision	Proved lower bound for $\lvert F\rvert$
$[0, 8]$	—	exact estimates	$1/120000$
$[7, 41]$	272	256 bits	$9/5000000$
$[40, 100]$	510	256 bits	10^{-16}
$[100, 104]$	32	192 bits	$4 \cdot 10^{-16}$

The 510 cells comprise 160, 170, and 180 cells in $[40, 60]$, $[60, 80]$, and $[80, 100]$. Their exact norm floor is

$$\frac{22271745300542300149807}{850705917302346158658436518579420528 64} > 10^{-16}.$$

Exact rational cover checks include the spatial subdivisions, all shared boundaries, and both time endpoints. The spatial overlaps $[7,8]$, $[40,41]$, and the shared boundary $x = 100$, together with evenness, prove (6.1).

The last band also certifies the centered-residual mechanism. On each of its 32 cells, freeze

$$a = \frac{\pi}{16}, \qquad b = \frac{1}{4}\log\frac{x_c + 2t_c a}{4\pi\sin(4a)}, \qquad \mu = a^2 + b^2, \qquad C_j = -h_{j+2} - 2ah_{j+1} - \mu h_j.$$

The complete even source is shifted before theta indices are separated. With eight retained indices, integration cutoff $3/2$, spatial order 12 and time order 8, the computation encloses all center jets through order 28 and complete derivative majorants through order 29. Centered coefficients are combined before Taylor transport, with the parameters frozen on the cell. The cell-constant normalization $e^{\pi x_c/8}$ rescales each physical jet equally. For normalized enclosures P_0, P_1, Q_0, Q_1 of h_0, h_1, C_0, C_1, the actual decisions are

$$E_{\mathrm{low}} = \operatorname{dist}(0, P_0)^2 + \operatorname{dist}(0, P_1)^2 > 0, \qquad E_{\mathrm{low}}/9 - \sup|Q_0|^2 - \sup|Q_1|^2 > 0.$$

Thus $C_0^2 + C_1^2 \le E/9$ and $\mu \ge 2/5$ throughout every cell. The exact stationary identity $\mathcal{L}_j = \mu h_j^2 + h_j C_j$ at $h_{j+1} = 0$ yields

$$\mathcal{L}_0 \ge E/15 \quad (h_1 = 0), \qquad \mathcal{L}_1 \ge -5E/72 \quad (h_2 = 0).$$

The second strict stationary sign in Theorem 6.1 is a separate validated signed decision; it is stronger than the bounded-loss conclusion supplied by this cone. At each level the band has 24 derivative-exclusion cells and eight strict signed cells, with no claim that a signed cell alone proves existence or uniqueness of a stationary branch.

Appendix B identifies the retained drivers, records, exact auditors, and earlier independent internal replays. In particular, the higher-height auditor checks 6682 assertions and the centered-band auditor checks 547. Those checks concern arithmetic, decisions, complete coverage, and source bindings. The complete-source contour bounds used for the latest band are proved in Appendix A. The computations establish the stated closed bounded region; they do not extend it to arbitrary heights.

7 Obstructions and the remaining signed estimate

The outer theorem and the finite cover give complementary information. For any fixed $\varepsilon > 0$, the unresolved positive-time domain is contained in the finite transition strip $104 < |x| < X_\varepsilon$ and the high-height inner region

$$0 < t \le 1/2, \quad |x| \ge X_\varepsilon, \quad t\log(|x|/(4\pi)) < 4 + \varepsilon. \tag{7.1}$$

The first missing quantitative task is to evaluate a usable X_ε. The second is an arithmetic sign estimate in (7.1). Several tempting stronger statements can be ruled out before pursuing that estimate.

7.1 Absolute first-channel dominance fails below four

At the exact common-carrier coordinate v, let $M = \lfloor e^{2v} \rfloor$. For the indices $\lceil M/2 \rceil \le n \le M$, we have

$$td_n^2 - 2td_n v = t[(d_n - v)^2 - v^2] \ge -tv^2.$$

Consequently

$$\sum_{n\le e^{2v}} |A_n| \ge (M/2-1)M^{-1/2}e^{-tv^2}. \tag{7.2}$$

With fixed a, $v=b_1$, and $t=\vartheta/\log x$ for fixed $0\le\vartheta<4$, the right side grows as a positive constant times $x^{1/4-\vartheta/16}$. The first weight has modulus one. Thus the remaining channels are not a small absolute perturbation of the first at this point of the carrier. The estimate leaves cancellation within the sum and after integration available.

7.2 The pointwise integrand need not have the desired sign

For a fixed $v=b$, write

$$D_j=\sum_{n\le e^{2b}} A_n(t,x;b)[-a+i(b-d_n)]^j, \qquad p_j=\Re(UD_j),$$

where U is the unit phase of $e^{t(b+ia)^2}\phi_1(b+ia)e^{ixb}$. Define

$$\mathfrak{H}_j=|D_{j+1}|^2-\Re(D_j\overline{D_{j+2}}), \qquad \mathfrak{W}_j=D_{j+1}^2-D_jD_{j+2}.$$

The elementary real-part identity gives

$$2(p_{j+1}^2-p_jp_{j+2})=\mathfrak{H}_j+\Re(U^2\mathfrak{W}_j). \tag{7.3}$$

A sufficient condition independent of phase would be $\mathfrak{H}_j\ge|\mathfrak{W}_j|$. The exact common-carrier sums, enclosed with Arb/Acb arithmetic, reject that stronger condition:

x	t	$\mathfrak{H}_0-\lvert\mathfrak{W}_0\rvert$ (approximate value)
211	$1/\log 211$	-0.0277198
4898.7	$2/\log 4898.7$	-1.26105

Here $a=\pi/16$ and $b=b_1$ from (5.13). At the second point, the actual carrier phase also gives $p_1^2-p_0p_2\approx-0.01099197<0$. The retained record encloses these values strictly below zero; Appendix B identifies the replay. These are pointwise integrand forms. They do not equal the Laguerre forms after integration, whose products include cross terms between different carrier coordinates as well as different indices. A proof based on the pointwise positivity of this integrand cannot cover those points. Their convention is $t\log x=1$ or 2, placing them outside Theorem 1.1.

The staircase endpoints also carry structure. When index n enters at $v=d_n$, direct substitution in (5.10) gives

$$e^{t(d_n+ia)^2}\phi_1(d_n+ia)e^{ixd_n}A_n(t,x;d_n)(-a)^j=e^{-ta^2}\phi_n(ia)(-a)^j. \tag{7.4}$$

The oscillatory arithmetic factor cancels at this boundary. Replacing the staircase by a fixed sum and ignoring these boundary terms would lose part of the complete-source identity.

7.3 A small residual norm can impose an unnecessary condition

Consider the real-rooted backward-heat control $P(t,y)=y^2-d^2-2t$, with $d\ge0$ and $t>0$. Its Laguerre forms are $2y^2+2d^2+4t>0$ and 4. At $y=0$, put $D=d^2+2t$. Then $P=-D$, $P_y=0$,

$E = D^2$, and the centered residuals are $C_0 = -2 + \mu D$, $C_1 = -4a$. The cone $C_0^2 + C_1^2 \le \eta^2 E$ would require

$$\eta \ge \frac{\sqrt{(2-\mu D)^2 + 16a^2}}{D}. \tag{7.5}$$

For bounded prescribed a, μ, this can become large as the pair closes even though the required negative loss is zero. This is a control example, not a statement that the theta cone fails. It shows why favorable curvature should not be paid as an adverse error. Growth as $t \downarrow 0$ is also compatible with local boundedness at every positive time.

The focused remaining target is therefore the one-sided estimate

$$\boxed{h_j C_j \ge -\mu h_j^2 - K_j E \quad \text{on } h_{j+1} = 0, \qquad j = 0, 1,} \tag{7.6}$$

where $K_j \ge 0$ is locally bounded through every candidate positive-time collision. By (5.3), this is exactly the sufficient stationary inequality $\mathcal{L}_j \ge -K_j E$. In particular, K_j may not be defined by an uncontrolled quotient with the unknown energy in its denominator. All arithmetic cross terms and complete source errors remain part of the estimate.

A natural next analytic step is to replace the absolute majorant in (2.11) by bounds for oscillatory arithmetic blocks and their logarithmic moments. This may lower the sufficient threshold four, but such an improvement has not been proved here. The octonionic formulation continues to identify the required orientation; the new content must come from the theta-specific arithmetic estimate.

A Complete source and integration error bounds

The following bounds make the numerical and shared-source constructions independent of an unpaid finite theta truncation. Set

$$S_k(q,d) = \sum_{r=0}^{k} \binom{k}{r} \frac{d^{k-r} r!}{q^{r+1}} = \int_0^\infty (v+d)^k e^{-qv}\, dv \qquad (q > 0).$$

Let $0 \le t \le \tau \le 1/2$, $0 < a < \pi/8$, and $x \ge 0$. For an index cutoff N, put $m = N + 1$ and

$$c_a = \cos 4a, \quad \alpha = \pi m^2, \quad \beta = c_a \alpha, \quad q_N = 4\beta - 9, \quad \rho_N = ((m+1)/m)^4 e^{-\pi c_a (2m+1)}.$$

If $\rho_N < 1$, $q_N > 0$, and $8\beta \ge \tau$, the complete omitted index contribution in (5.1) is at most

$$E_{N,k} = e^{-ax} \frac{(2\alpha^2 + 3\alpha) e^{-\beta}}{1 - \rho_N} S_k(q_N, a). \tag{A.1}$$

For a cutoff $U \ge 0$, put

$$\beta_U = \pi c_a e^{4U}, \quad \rho_U = 16 e^{-3\beta_U}, \quad q_U = 4\beta_U - 9 - 2\tau U, \quad B_U = \tau U^2 + 9U - \beta_U.$$

If $\rho_U < 1$, $q_U > 0$, and $8\beta_U \ge \tau$, the entire source on $u \ge U$ contributes at most

$$E_{U,k} = e^{-ax} \frac{(2\pi^2 + 3\pi) e^{B_U}}{1 - \rho_U} S_k(q_U, U + a). \tag{A.2}$$

Thus $E_{N,k} + E_{U,k}$ pays the omitted domain of an N-term integral on $[0, U]$. Validated retained quadrature and cell variation are additional errors.

To prove the bounds, use

$$|\phi_n(u+ia)| \le (2\alpha_n^2+3\alpha_n)e^{9u-c_a\alpha_n e^{4u}}, \quad |e^{t(u+ia)^2}| \le e^{\tau u^2}, \quad |u+ia|^k \le (u+a)^k.$$

For $n \ge m$, successive coefficient-majorant ratios are bounded by ρ_N uniformly for $u \ge 0$. The inequality $e^{4u} \ge 1+4u+8u^2$ gives

$$\tau u^2+9u-\beta e^{4u} \le -\beta - q_N u - (8\beta-\tau)u^2 \le -\beta - q_N u.$$

Summing the full geometric index bound and integrating proves (A.1). For $u \ge U$, the exponent $\tau u^2+9u-\pi c_a e^{4u}$ is concave and has slope $-q_U$ at U. Its tangent bound and the geometric ratio ρ_U give (A.2). Adding the bounds overcounts their intersection, which is harmless.

For the centered residual C_j, replace any factor $S_j(q,d)$ in these payments by $S_{j+2}(q,d)+b^2S_j(q,d)$, since $|u^2-b^2| \le (u+a)^2+b^2$. For the shared carrier (5.11), set

$$\widehat{\beta}_V = \pi\cos(4a)e^{4V}, \quad \widehat{q}_V = 4\widehat{\beta}_V - 10 - 2\tau V, \quad \widehat{B}_V = \tau V^2+10V-\widehat{\beta}_V.$$

If $\widehat{q}_V > 0$ and $8\widehat{\beta}_V \ge \tau$, the majorant (5.12) and the same tangent argument give the full tail payment

$$e^{-ax}2(2\pi^2+3\pi)e^{\widehat{B}_V}S_j(\widehat{q}_V, V+a). \tag{A.3}$$

The same degree-two replacement pays the shared expression for C_j, because $0 \le v-d_n \le v$.

For completeness, these absolute errors can be made small at the gamma exponential scale at arbitrary height. For $x \ge 41$, $p > 0$, and $\ell_x = \log x$, take

$$a_x = \pi/8 - 1/x, \quad m = \left\lceil \sqrt{\frac{x}{2\pi}(p\ell_x+32)} \right\rceil, \quad N = m-1, \quad U = \frac{\ell_x + 2\log(\ell_x+8)}{4}.$$

All preceding payment hypotheses hold, and for fixed k,

$$e^{\pi x/8}E_{N,k} = O_{p,k}(x^{5/2-p}(\log x)^{3/2}), \qquad e^{\pi x/8}E_{U,k} = O_{k,A}(x^{-A}) \quad (A>0). \tag{A.4}$$

Indeed $c_a = \sin(4/x) \ge 2/x$, $\beta \ge p\ell_x+32$, and $m^2 \ge 2x/\pi$. Hence $-\log\rho_N \ge 4\pi m/x - 4/m \ge 2\pi m/x$ and $(1-\rho_N)^{-1} \le 1+x/(2\pi m)$. Together with $e^{-a_x x} = e\,e^{-\pi x/8}$, substitution gives the first bound. For the second, $\beta_U \ge 2\pi(\ell_x+8)^2$, $U \le \ell_x/2+2$, and $B_U \le -47\ell_x^2/8 - 181\ell_x/2 - 364$. These are absolute error estimates. They neither establish a norm gap nor replace a signed estimate of the retained source. On a spatial cell, freeze the contour and cutoffs before differentiating or transporting physical jets.

B Reproducibility and scope of the checks

All paths in this appendix are relative to `papers/investigations-2/active/21_quaternion_theta_geometry/` in the accompanying repository. The analytic proofs above are distinct from the finite interval decisions. The latter use complete error bounds, outward rounding, and closed-cell coverage; hashes identify the evaluated sources and records, but do not substitute for their mathematical content.

The bounded results can be regenerated using the drivers `numerics/certify_compensated_stationary_time_extension.py`, `numerics/certify_correlated_source_height_extension.py`, and `numerics/certify_phase_centered_residual.py`. The correlated-height guide

`numerics/CORRELATED_SOURCE_HEIGHT_README_20261014.md` gives the parameters for its three bands. The full-time certificate is `numerics/COMPENSATED_STATIONARY_TIME_EXTENSION_20261014.json`; the final centered-band record is `numerics/THETA_CENTERED_RESIDUAL_100_104_20261014.json`. The independent internal replays are documented in Reviews 6, 9, and 11 in `reviews/`. They reproduced the numerical records after removal of runtime metadata.
For compact exact audits, run the following from `numerics/`:

```
python3 audit_correlated_source_height_records.py
python3 audit_theta_centered_residual_record.py
python3 check_octonion_jet_identities.py
python3 check_theta_centered_residual_identities.py
```

The centered-band auditor additionally requires `-record THETA_CENTERED_RESIDUAL_100_104_20261014.json`. Each requires the option `-output` followed by a desired JSON path. The first two inspect the retained records and bound source versions; the latter two check 248 and nine exact algebraic identities, respectively. The octonionic identities, centered residual, and outer analytic argument have separate internal assessments in Reviews 10–12. The diagnostic `numerics/check_outer_cusp_and_phase_obstructions.py`, also run with `-output`, requires python-flint and imports `numerics/check_shifted_contour_payments.py`. Its retained record `numerics/OUTER_CUSP_AND_PHASE_OBSTRUCTION_CHECKS_20261014.json` contains the pointwise counterexamples in Section 7, twelve staircase boundary checks, exact four-jet perturbation algebra, and selected complete-source point enclosures. It does not certify a new rectangle or prove the uniform asymptotic theorem.
The continuation is governed by two distinct obligations: making the outer threshold effective and obtaining the one-sided inner-region bound (7.6). A fixed-time asymptotic, a selected-point calculation, or a relative error divided by an unknown physical norm cannot replace either obligation.

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